



Faculty of Science and Technology

Programme Structure & Syllabus

For

Second Year

Master of Science Computer Science

(MSC CS)

P-2024

With Effect from Academic Year 2025– 2026

	Authority	Date
Proposed by	Board of Studies in Computer Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

1. Chairman – BoS

Dean of Faculty

Director, IQAC

Title: Programme Structure and Syllabus

Form No: IQAC-101

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Science	
Emerge as renowned research and innovation-oriented department in the field of Information Technology	
Mission of the Department of Computer Science	
M1	To impart industry centric education through research and innovation
M2	To develop skilled professionals to meet global socio-economic challenges
M3	To introduce best practices in software development adhering to moral and ethics.
M4	To foster collaborative approach and leadership skills.
Values of the Department of Computer Science	
Integrity	Brightness
Innovation	Diversity
Ethics	Competitive
Leadership	Lucidity

2. Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	Post Graduates will have a thorough foundational knowledge of the principles and practices of Computer Science, and be proficient in applying them. (Based on Knowledge)
PEO2	To be able to adapt new tools/techniques to effectively contribute as an individual and lead the software development team to provide economically and technically viable solutions to a problem. (Based on Skills)
PEO3	Commit to ethical and moral practices for providing solutions to societal problems through life-long learning (Based on Attitude)

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3
M1	3	2	3
M2	1	3	2
M3			
M4	1	2	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Outcomes (POs)

PO No.	Statement
PO1	Possess profound knowledge of theory and principles of basic mathematics, statistical techniques and algorithms in the field of computer science.
PO2	Ability to think critically and select economically and technically viable solutions to a given problem.
PO3	Ability to design, develop and deploy solutions for identified problems from diverse domains.
PO4	Provide proof of concept and present novel solutions to challenges encountered by industry.
PO5	Adopt new programming languages, tools, and processes followed in industry
PO6	To work independently or as an effective team member of a software development project.
PO7	Effectively communicate with diverse clients and stakeholders for software project management.
PO8	Interpersonal skills to interact with people from diverse cultural backgrounds
PO9	Adhere to the professional, ethical and legal practices followed in the IT industry.
PO10	Possess software project management skills like: cost estimation, effort estimation, scheduling, and team building.

Mapping of PEOs to Pos

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
PEO1	3	2	3	2	1	-	-	-	-	-
PEO2	1	3	-	2	3	2	2	1	1	2
PEO3	-	-	2	-	-	1	1	2	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

4. Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Choose appropriate data structure, algorithm, software development paradigm to address real life problems.
PSO2	Apply modern tools and techniques used in Data Science and Game Development.
PSO3	Provide the solutions to computing challenges from multidisciplinary fields.
PSO4	Possess a research and development attitude focusing on knowledge creation and innovation.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3	PSO4
PEO1	-	-	-	4
PEO2	2	2	1	-
PEO3	3	-	3	-
PEO4	-	-	-	4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

SYMSCCS Semester –III									
Course Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CIE	ESE	Total
DSC	MSCCS24301	Distributed Database Concepts	3	0	0	3	60	40	100
DSC	MSCCS24302	Distributed Database Concepts Lab	0	0	2	1	15	10	25
DSC	MSCCS24303	Digital Image Processing	3	0	0	3	60	40	100
DSC	MSCCS24304	Digital Image Processing Lab	0	0	2	1	15	10	25
DSC	MSCCS24305	Soft Computing	2	0	0	2	50	0	50
DSC	MSCCS24306	Data Compression	2	0	0	2	50	0	50
DSE	MSCCS24307WD	C# .NET Programming	3	0	0	3	60	40	100
	MSCCS24307DS	Machine Learning							
DSE	MSCCS24308WD	C# .NET Programming Lab	0	0	2	1	15	10	25
	MSCCS24308DS	Machine Learning Lab							
RP	MSCCS24309	Research Project-I	0	0	8	4	60	40	100
		Total	13	0	14	20	385	190	575

Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit

MSCCS24301 : Distributed Database Concepts

Course Type	DSC	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hrs./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	-	Total Marks	100
Total	3 Hrs./Week	Total	3		

Course Description

This course provides students with an in-depth understanding of distributed database systems, emphasizing industry-aligned practices and emerging trends. It covers foundational concepts such as architecture, design strategies, and integrity control, alongside advanced topics like query optimization, distributed transactions, and fault-tolerant systems. Additionally, students will explore modern industry technologies like NoSQL, distributed storage systems, and cloud databases, preparing them to design and manage scalable, reliable, and high-performance distributed systems.

Course Outcomes

CO No.	Statement
1	Explain distributed database architectures, models.
2	Design distributed databases with consideration for fragmentation, allocation, and integrity constraints.
3	Apply query decomposition, optimization techniques, and modern query processing algorithms.
4	Implement and analyze concurrency control and distributed transaction mechanisms in real-world scenarios.
5	Evaluate the reliability, fault tolerance, and scalability of distributed database systems using modern tools and frameworks.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs			BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	
CO1	3	2	2	1										2
CO2	3	3	3		2						2			4
CO3		3	3	2	2									4
CO4		2	3	3	2									4
CO5	3	3	3	3	3									4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Unit 1	Distributed databases: An overview	09 hours	CO	BTL
Features of distributed vs centralized databases; Benefits of distributed databases in modern applications; Promises and problem areas of DDBMS; DDBMS architectural models; Distributed catalog management; Emerging trends in distributed databases, including microservices-based architectures.			1	2
Unit 2	Distributed database design & query processing	09 hours	2	2

Alternative design strategies; Distributed design issues; Fragmentation and allocation strategies; Concepts of join graphs; Query processing challenges; Objectives of query processing; Complexity of relational algebra; Query processors; Role of cloud-based query processing platforms.				
Unit 3	Query decomposition, data localization & Optimization of distributed queries	09 hours	3	4
Query decomposition; Data localization; Query optimization; Centralized vs distributed optimization; Join ordering in fragment queries; Introduction to NoSQL databases (MongoDB); Optimization techniques for NoSQL queries; Performance tuning in distributed environments.				
Unit 4	Management of distributed transactions	09 hours	4	4
Framework for transaction management; Supporting atomicity; Concurrency control mechanisms (locking, timestamp ordering); Distributed deadlocks and prevention; Optimistic concurrency control methods; Architectural considerations for scaling distributed transactions.				
Unit 5	Distributed DBMS reliability	09 hours	5	4
Reliability measures; Fault tolerance in distributed systems; Local and distributed reliability protocols; Handling site failures and network partitioning; Introduction to cloud databases (AWS DynamoDB, Google Spanner); Fault tolerance in cloud-native distributed systems; Disaster recovery and high availability in distributed environments.				

Reference books:

1	Database System Concepts, Abraham Silberschatz, Henry Korth, and S. Sudarshan, McGraw-Hill, 7th Edition, 2020
2	Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, Pearson, 7th Edition, 2016

Textbooks:

1	Distributed Database Systems, Stefano Ceri and Giuseppe Pelagatti, McGraw-Hill, 2nd Edition, 2008
2	Principles of Distributed Database Systems, M. Tamer Özsu and Patrick Valduriez, Springer, 3rd Edition, 2011

Weblinks:

1	www.mongodb.com/docs/manual
2	https://www.geeksforgeeks.org/distributed-database-system
3	https://www.tutorialspoint.com/distributed_dbms/index.htm
4	https://www.javatpoint.com/distributed-database

MSCCS24302 : Distributed Database Concepts Lab

Course Type	DSC	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hrs./Week	Lecture	-	CIE Marks	15
Tutorial	0 Hr./Week	Tutorial	-	ESE Marks	10
Practical/Studio	0 Hr./Week	Practical/Studio	1	Total Marks	25
Total	2 Hrs./Week	Total	1		

Course Description

This course provides students with an in-depth understanding of distributed database systems, emphasizing industry-aligned practices and emerging trends. It covers foundational concepts such as architecture, design strategies, and integrity control, alongside advanced topics like query optimization, distributed transactions, and fault-tolerant systems. Additionally, students will explore modern industry technologies like NoSQL, distributed storage systems, and cloud databases, preparing them to design and manage scalable, reliable, and high-performance distributed systems.

Course Outcomes

CO No.	Statement
1	Explain distributed database architectures, models.
2	Design distributed databases with consideration for fragmentation, allocation, and integrity constraints.
3	Apply query decomposition, optimization techniques, and modern query processing algorithms.
4	Implement and analyze concurrency control and distributed transaction mechanisms in real-world scenarios.
5	Evaluate the reliability, fault tolerance, and scalability of distributed database systems using modern tools and frameworks.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs			BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	
CO1	3	2	2	1										2
CO2	3	3	3		2						2			4
CO3		3	3	2	2									4
CO4		2	3	3	2									4
CO5	3	3	3	3	3									4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

List of Experiments:

Sr. No.	Lab Assignment	CO	BTL
1	Install MongoDB and create a database with a collection. Perform basic CRUD operations.	1	3
2	Create an aggregate data model in MongoDB for an online store. Query product reviews from the model.	1	4
3	Implement sharding in MongoDB. Set up multiple shards and observe data distribution across shards.	2	4
4	Implement version control in a MongoDB collection using a version field. Track document updates.	2	4
5	Perform a Map-Reduce operation in MongoDB to calculate total sales for each product category.	3	5
6	Create a schema-less collection in MongoDB. Insert documents with varying structures and query them.	3	4
7	Use CouchDB to create a document database for a blog. Insert posts and query them by tags or authors.	4	4
8	Install MySQL, create a database and table, and perform CRUD operations using SQL queries.	1	3
9	Design a relational database schema for a library management system. Query book transactions.	2	4
10	Use MySQL aggregate functions (SUM, COUNT, AVG) to analyze a sales dataset.	2	3
11	Create a partitioned table in MySQL. Partition student records by graduation year and query them.	3	4
12	Demonstrate the ACID properties in MySQL by performing dependent updates with COMMIT and ROLLBACK.	4	4
13	Design a relational database with customers and orders. Use foreign keys and JOIN queries.	3	4
14	Use PostgreSQL to create a table with JSON data. Insert and query JSON documents.	4	4
15	Create a simple distributed database simulation by dividing data into logical partitions. Query data.	5	5

Reference books:

1	Database System Concepts, Abraham Silberschatz, Henry Korth, and S. Sudarshan, McGraw-Hill, 7th Edition, 2020
2	Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, Pearson, 7th Edition, 2016

Textbooks:

1	Distributed Database Systems, Stefano Ceri and Giuseppe Pelagatti, McGraw-Hill, 2nd Edition, 2008
2	Principles of Distributed Database Systems, M. Tamer Özsu and Patrick Valduriez, Springer, 3rd Edition, 2011

Weblinks:

1	www.mongodb.com/docs/manual
2	https://www.geeksforgeeks.org/distributed-database-system
3	https://www.tutorialspoint.com/distributed_dbms/index.htm
4	https://www.javatpoint.com/distributed-database

MSCCS24303 :: Digital Image Processing

Course Type	DSC	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	100
Total	3 Hr./Week	Total	3		

Course Description

The Digital Image Processing course for the Master of Computer Science program provides a comprehensive understanding of image processing techniques, industry applications, and project-based learning. It covers fundamental concepts, including image representation, enhancement, restoration, segmentation, feature extraction, compression, and object recognition. Advanced topics such as real-time image processing applications, AI/ML-based approaches, and emerging trends are explored. The course follows an industry-oriented approach, integrating hands-on mini-projects to equip students with practical skills for research and professional applications. By the end of the course, students will be able to design and implement image processing solutions for real-world challenges in domains like medical imaging, surveillance, remote sensing, and smart automation.

Course Outcomes

CO No.	Statement
1	Recall fundamental concepts, techniques, and applications of digital image processing.
2	Explain image enhancement, restoration, and filtering techniques for improving image quality.
3	Implement segmentation, edge detection, and feature extraction methods for object identification.
4	Compare different image compression and object recognition techniques for optimized storage and retrieval.
5	Assess real-time image processing applications and develop solutions using AI and ML-based approaches.
6	Design and develop an industry-oriented mini project for real-world image processing applications.

Mapping of COs to POs and PSOs

CO No.	Pos										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	PSO 4	
CO 1	1										3	2			1,2
CO 2		2	3								2	1			2,4
CO 3			3	1	1						2	1			1,3,4
CO 4			2	1	1						1	3			5,6
CO		2	3	2	2	1					2	3	4		4,5

5															
CO 6			3	3	3	2	1	1		2	3	4	5	6	6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No. 1	Introduction to DIP	6 Hours	CO	BTL
	Basics of Digital Image Processing; Applications: Medical Imaging, Agriculture, Satellite Imaging, AI-based Imaging, Image Representation: Pixels, Resolution, Color Models (RGB, CMY, HSV, Grayscale); Image Acquisition and Sampling; Relationship Between Pixels (Adjacency, Connectivity)		1	1,2
Unit No. 2	Image Enhancement and Restoration	8 Hours		
	Spatial Domain Methods: Histogram Equalization, Contrast Stretching; Frequency Domain Methods: Fourier Transform, Low/High Pass Filters; Noise Removal Techniques: Gaussian, Salt & Pepper, Median Filtering; Restoration Techniques: Inverse Filtering, Wiener Filtering		2	2,4
Unit No. 3	Image segmentation & Feature Extraction	7 Hours		
	Edge Detection: Sobel, Prewitt, Canny; Region-Based Segmentation: Thresholding, Watershed Algorithm; Morphological Image Processing: Dilation, Erosion, Opening, Closing; Feature Extraction: Contours, Shape Analysis		3	1,3,4
Unit No. 4	Image Compression & Object Recognition	7 Hours		
	Lossless & Lossy Compression: Huffman Coding, JPEG, Wavelet Compression; Object Detection & Recognition Techniques; Template Matching, Feature Matching; Introduction to AI & Deep Learning in Image Processing		4	5,6
Unit No. 5	Image Processing in Real-Time Applications	8 Hours		
	Image Processing in IoT and Smart Systems; Real-Time Video Processing (OpenCV Introduction); Motion Detection & Tracking Algorithms; Augmented Reality (AR) & Virtual Reality (VR) Basics		5	4,5
Unit No. 6	Project-Based Learning & Emerging Trends	9 Hours		
	Case Study: Real-World Applications (Medical Imaging, Industrial Automation, Security); AI & ML-Based Image Processing Trends; Ethical Considerations in Image Processing; Mini Project: Implement a basic DIP application (e.g., face detection, smart farming analysis, medical diagnosis system)		6	6

Textbooks

1	<i>Fundamentals of Image Processing, Anil K. Jain, Prentice Hall Information and System Sciences Series, 1989.</i>
2	<i>Digital Image Processing and Analysis, B. Chanda, D. Dutta Mujumder, Prentice Hall India Learning Private Limited; 2nd edition 2011.</i>
3	<i>Digital Image Processing, Jayaraman S, Veerakumar T, Esakkirajan S, McGraw Hill Education; 1st edition, 2017.</i>
4	<i>Digital Image Processing, S. Sridhar, Oxford University Press; Second edition, 2016.</i>
5	<i>Digital Image Processing, Rafael C. Gonzalez, Richard E. Wood; Pearson Education; Fourth edition, 2018.</i>
6	<i>Digital Image Processing Using Matlab, Rafael C. Gonzalez, Richard E. Wood, Steven L. Eddins;</i>

Reference Books / Journal Articles / Weblink

1	<i>Digital Image Processing</i> , Rafael C. Gonzalez, Richard E. Wood; Pearson Education; Third Edition, 2007.
2	<i>Fundamentals of Digital Image and Video Processing</i> , https://www.coursera.org/learn/digital , (accessed on 21/3/2025)
3	<i>Principles of Digital Image Processing – Advanced Methods</i> , https://imagingbook.com/wp-content/uploads/2013/06/burgerburgeuticsvolc_extras.pdf , (accessed on 21/3/2025)

MSCCS24304 :: Digital Image Processing Lab

Course Type	DSC	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Practical/Studio	2 Hr./Week	Practical/Studio	1	CIE Marks	15
				ESE Marks	10
				Total Marks	25

Course Outcomes

CO No.	Statement
1	Understand the Basics of Digital Image Processing.
2	Apply Image Enhancement & Restoration Techniques.
3	Analyze and Implement Image Segmentation & Feature Extraction.
4	Apply Compression and Object Recognition Techniques.
5	Develop Real-Time Image Processing Applications.
6	Design and Implement an Industry-Oriented Image Processing Solution.

Mapping of COs to POs and PSOs

CO No.	Pos										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	PSO 4	
CO 1	3										2	2		1,2	2
CO 2		2	3	1							2	1		2,4	3
CO 3			3	2	1						1	1		1,3,4	4
CO 4			2	3	1						1	3		5,6	3
CO 5		3	3	2	3	2					3	2	1	3,5,6	5
CO 6	2	3	3	3	3	3	2	1	2	2	3	3	2	3,4,6	6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content**Instructions:**

To carry out the lab sessions for the Digital Image Processing course, students should use Python programming language along with Jupyter Notebook or Google Colab as the development environment for easy visualization and experimentation. The primary libraries required include OpenCV (cv2) for image processing tasks, NumPy for numerical operations, Matplotlib for image display and plotting, and scikit-image for advanced segmentation and filtering techniques. For deep learning applications, TensorFlow and Keras will be used to train and test object recognition models. Additionally, Pillow can be used for basic image handling, and PyWavelets for wavelet transforms. To handle compression and encoding, imageio and bitstring may also be included. All tools and libraries

are open-source and can be installed using pip, making the setup student-friendly and industry-relevant.

List of Experiments:

Sr. No	Name of Experiment	CO	BTL
1	Load and display an image in different color models (RGB, CMY, HSV, Grayscale)..	1	2
2	Perform image downsampling and upsampling, analyzing the effects on resolution.	1	2
3	Implement pixel adjacency and connectivity to identify connected components in an image.	1	3
4	Apply histogram equalization and contrast stretching on low-contrast images.	2	3
5	Perform image filtering using Fourier Transform (Low-Pass & High-Pass Filtering).	2	3
6	Implement Gaussian, Median, and Salt & Pepper noise removal on a noisy image.	2	3
7	Use Inverse and Wiener filtering to restore degraded images.	2	3
8	Implement Sobel, Prewitt, and Canny edge detection on sample images.	3	4
9	Perform region-based segmentation using thresholding and the Watershed Algorithm.	3	4
10	Apply dilation, erosion, opening, and closing operations on binary images.	3	3
11	Implement Huffman Coding and JPEG compression on an image.	4	3
12	Detect objects in an image using template matching and feature-based methods (ORB/SIFT).	4	3
13	Train and test a simple deep learning model for object recognition using OpenCV and TensorFlow.	4	4
14	Capture and display real-time video frames using OpenCV.	5	3
15	Implement motion detection and object tracking in a video feed.	5	5
16	Overlay virtual objects on a real-world scene using basic AR techniques.	5	5
17	Develop a basic DIP application (e.g., face detection, smart farming analysis, or a medical diagnosis system, etc.).	6	6

Textbooks

1	<i>Image Processing Using Python, Chityala, Ravishankar and Pudipeddi, Sridevi; CRC Press; Second Edition, 2020</i>
2	<i>Digital Image Processing, Rafael C. Gonzalez and Richard E. Woods, Fourth Edition, 2018.</i>
3	<i>Hands-On Image Processing with Python, Sandipan Dey, Packt Publishing, 2018.</i>
4	<i>Digital Image Processing, S. Sridhar, Oxford University Press; Second edition, 2016.</i>
5	<i>Fundamentals of Digital Image Processing, Anil K. Jain, Prentice Hall, 1989.</i>

Reference Books / Journal Articles / Databases Documents of Professional Bodies (ASME/ASTEM, etc)

1	<i>Image Processing with Python A practical approach, Manjushree Nayak, Devanand Bhonsle, Shruti Tiwari, Roshni Rahangdale, IOP Publishing Ltd., 2024.</i>
2	<i>Detection and Counting of Microaneurysm for Early Diagnosis of Maculopathy, Chetan Pattebahadur, Ramesh Manza, Anupriya Kamble, Priyanka Verma; Springer, Singapore, 2020</i>
3	<i>Design a Novel Detection for Maculopathy Using Weightage KNN Classification, Chetan Pattebahadur,</i>

	Ramesh Manza, Anupriya Kamble, Springer Nature Singapore Pte Ltd., 2019
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Digital Resource

- | | |
|---|---|
| 1 | https://www.mygreatlearning.com/academy/learn-for-free/courses/digital-image-processing , accessed on Friday, March 28, 2025. |
| 2 | https://opencv.org/university/free-opencv-course/ , accessed on Friday, March 28, 2025. |
| 3 | https://www.coursera.org/learn/introduction-computer-vision-watson-opencv , accessed on Friday, March 28, 2025. |

MSCCS24305: Soft Computing

Course Type	DSC	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	02 Hr./Week	Lecture	2	CIE Marks	50
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	00
Practical/Studio	00 Hr./Week	Practical/Studio	0	Total Marks	50
Total	02 Hr./Week	Total	2		

Course Objectives

The objective of this course is to familiarize students with soft computing concepts and techniques. It aims to help students understand and apply neural networks, fuzzy logic, genetic algorithms, and hybrid systems for problem-solving in various applications.

Course Outcomes

CO No.	Statement
1	Learn soft computing techniques and their role in problem-solving.
2	Analyze various artificial neural network techniques.
3	Define and apply fuzzy systems for problem-solving.
4	Apply genetic algorithms to solve problems.
5	Identify suitable soft computing methods for various applications.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3	2	3	2	1	2	2	2	1	2	3	2	3	2	1, 2,3
CO2	3	3	3	2	1	2	2	1	1	2	3	2	2	2	3,4
CO3	2	2	3	3	1	2	2	1	1	2	2	2	3	3	3,4
CO4	3	3	2	3	2	3	2	1	2	3	3	3	3	3	4,5
CO5	3	2	3	2	1	3	3	2	2	2	3	3	3	3	3,4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

			CO	BTL
Unit No – I	Title of Unit - Introduction to Soft Computing	Hours – 06	1	1,2,3
- Basic concepts of Soft Computing - Historical Development, Definitions, Characteristics - Advantages and Disadvantages - Components of Soft Computing: - Fuzzy Logic and Computing - Neural Computing - Evolutionary Computing - Genetic Algorithms - Probabilistic Reasoning				
Unit No – II	Title of Unit – Artificial Neural Networks	Hours – 06	2	3,4
- Biological Neurons vs. Artificial Neurons - Artificial Neural Network Model - Neural Network Architectures - Neural Network Models: - Perceptron, Adaline, Madaline - Single-layer and Multi-layer Networks - Backpropagation Algorithm - Learning Methods: Hebbian, Competitive, Boltzmann, etc. - Competitive Learning Networks: - Kohonen's Self-Organizing Networks - Hopfield Networks				
Unit No – III	Title of Unit - Fuzzy Logic and Fuzzy Systems	Hours – 06	3	3,4
- Fuzzy Logic, Fuzzy Sets, and Operations - Fuzzy Relations, Arithmetic, and Measures - Fuzzy to Crisp Conversions: Lambda Cuts, Defuzzification Methods - Fuzzy Rules and Reasoning - Fuzzy Inference Systems: - Mamdani and Sugeno Fuzzy Models - Neuro-Fuzzy Modeling				

Unit No – IV	Title of Unit – Genetic Algorithms	Hours – 06	4	4,5
• Introduction to Genetic Algorithms • Encoding, Operators, Basic Genetic Algorithm • Crossover and Mutation • Multi-Objective Genetic Algorithm (MOGA) • Applications in Search, Optimization, and Pattern Recognition				
Unit No – V	Title of Unit – Applications of Soft Computing	Hours – 06	5	3,4
• Soft Computing in: ○ Pattern Recognition (Character Recognition) ○ Image Processing and Computer Vision ○ Mobile Ad-hoc Networks ○ Information Retrieval & Semantic Web ○ Software Engineering • Advanced Optimization Techniques: ○ Simulated Annealing ○ Tabu Search ○ Ant Colony Optimization (ACO) ○ Particle Swarm Optimization (PSO)				

Textbooks

1	Sivanandam, S. N., & Deepa, S. N. (2018). Principles of Soft Computing (2nd ed.). Wiley Publications.
2	Jang, J. S. R., Sun, C. T., & Mizutani, E. (1997). <i>Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence</i> . PHI.

Reference Books / Journal Articles / Weblink

1	Kumar, S. (2004). <i>Neural Networks: A Classroom Approach</i> . Tata McGraw-Hill.
2	Ross, T. J. (2017). <i>Fuzzy Logic with Engineering Applications</i> (3rd ed.). Wiley India.
3	https://onlinecourses.nptel.ac.in/noc20_cs17/preview

MSCCS24306 : Data Compression

Course Type	DSC	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hrs./Week	Lecture	2	CIE Marks	50
Tutorial	0 Hr./Week	Tutorial	-	ESE Marks	00
Practical/Studio	0 Hr./Week	Practical/Studio	-	Total Marks	50
Total	2 Hrs./Week	Total	2		

Course Description

This course introduces students to the principles, techniques, and applications of data compression. It covers the theoretical foundations, lossless and lossy compression algorithms, and the latest trends in the field. Students will learn to design, analyze, and implement data compression techniques, preparing them for research or industry applications in domains such as multimedia, storage, and networking.

Course Outcomes

CO No.	Statement
1	Explain the fundamentals of data compression and its importance in modern computing.
2	Implement and analyze lossless data compression algorithms.
3	Demonstrate the working of lossy compression techniques in multimedia.
4	Explore and implement advanced data compression tools and frameworks.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs			BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PSO 1	PSO 2	PSO 3	
CO1	3	2	2	1										2
CO2	3	3	3	2	2							2		4
CO3	3	3	2		2									4
CO4	2	3	3	3	3									4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit 1	Introduction to Data Compression	08 hours	CO	BTL
	Features and fundamentals of data compression; Lossy vs Lossless compression; Importance of data compression in multimedia and storage; Basic concepts of information theory – entropy, redundancy, and compressibility; Overview and classification of compression techniques.		1	2
Unit 2	Lossless Data Compression	07 hours	2	3

Entropy-based methods: Huffman coding, Arithmetic coding; Dictionary-based techniques: LZ77, LZ78, and LZW; Run-length encoding (RLE); Burrows-Wheeler Transform (BWT).				
Unit 3	Lossy Data Compression	07 hours	3	4

Transform-based methods: Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT); Image compression: JPEG, PNG, and WebP; Audio compression: MP3, AAC; Video compression: H.264/AVC, H.265/HEVC.				
Unit 4	Advanced Topics and Practical Applications	08 hours	4	4
Advanced methods: Fractal compression and neural network-based techniques; Practical tools: gzip, WinRAR, zlib (Python); Emerging trends: quantum data compression, Big Data, and IoT; Case studies and real-world applications.				

Reference books:

1	Data Compression: The Complete Reference, David Salomon, Springer, Volume N/A, 4th Edition, 2007
2	The Data Compression Book, Mark Nelson, Jean-Loup Gailly, M&T Books, Volume N/A, 2nd Edition, 1995

Textbooks:

1	Introduction to Data Compression, Khalid Sayood, Morgan Kaufmann, 5th Edition, 2022
2	Elements of Information Theory, Thomas M. Cover, Joy A. Thomas, Wiley-Interscience, 2nd Edition, 2006

Weblinks:

1	https://www.tutorialspoint.com/data_compression/index.htm
2	https://www.geeksforgeeks.org/data-compression-techniques
3	https://en.wikipedia.org/wiki/Data_compression

MSCCS24307WD : C#.NET Programming

Course Type	DSE	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	03 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	00 Hr./Week	Practical/Studio	0	Total Marks	100
Total	03 Hr./Week	Total	3		

Course Objectives

C#.NET Programming is a subject that focuses on the development of Windows and web applications using the C# programming language within the .NET framework. It covers the fundamentals of C# syntax, object-oriented programming concepts (such as classes, inheritance, and interfaces), and the use of .NET libraries for building robust, secure, and scalable applications. Topics often include Windows Forms, exception handling, file I/O, ADO.NET for database connectivity, and basic web development using ASP.NET.

Course Outcomes

CO No.	Statement
1	Understand the fundamentals of C# programming and the .NET framework.
2	Apply object-oriented programming principles using C# (such as encapsulation, inheritance, and polymorphism).
3	Develop Windows-based applications using Windows Forms and event-driven programming.
4	Implement data access using ADO.NET for CRUD operations in databases.
5	Build simple web applications using ASP.NET and manage web controls effectively.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3	2	3	2	1	2	2	2	1	2	3	2	3	2	1, 2,3
CO2	3	3	3	2	1	2	2	1	1	2	3	2	2	2	3,4
CO3	2	2	3	3	1	2	2	1	1	2	2	2	3	3	3,4

CO4	3	3	2	3	2	3	2	1	2	3	3	3	3	3	4,5
CO5	3	2	3	2	1	3	3	2	2	2	3	3	3	3	3,4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

			CO	BTL
Unit No – I	Title of Unit - Introduction :	Hours – 09	1	1,2,3
Basic Concepts, Features, Common Language Specification, C# Types: Simple type, Struct type, Object type Class type, Interfaces,String type, Arrays , Boxing & unboxing Conversions , Implicits , Explicits , Standard & User Defined Conversions.				
Unit No – II	Title of Unit – Control and Iterative Statements and Classes	Hours – 09	2	3,4
Control Statements : Selection Statements – if , Switch, Iteration Statements – For, For-Each, While , Do statements.				
Classes & Methods : Constructors & Destructors ,MethodsParameters, Overriding, Hiding class properties , Indexes , Modifiers, Class member Access, Multi cast deligates, Inheritance & Polymorphism : Inheritance- Basic class & Derived Class , Polymorphism , Base class with Virtual method, Derived class with override methods				
Unit No – III	Title of Unit – Interfaces and Exception handling	Hours – 09	3	3,4
Interfaces: Base, body , members , methods , properties , events, indexes, mapping, implementation				
Exception Handling : Checked & Unchecked statements, compiler, settings for overflow checking , Programmatic overflow checking , Exception handling statements – try & catch , try & finally , try- catch finally , throwing exception & rethrowing exception.				
Unit No – IV	Title of Unit – Basic of ASP.Net	Hours – 09	4	4,5
Web designing, web browser, web pages, home page, web site, web servers, world wide web , Concepts of hypertext, hypermedia, versions of HTML ,Evolution of .NET, Benefits of .NET Framework, Architecture of .NET Framework, Components of .NET Framework				
Unit No – V	Title of Unit – ASP.NET Page Life Cycle	Hours – 09	5	3,4

ASP.NET Page Life Cycle, understanding ASP.NET controls, applications, web servers, installation of IIS. Web forms, web form controls, server controls, client controls, adding controls to web form, buttons, text box, labels, checkbox, radio buttons, list box, drop, down list, Ad rotator control . Adding controls a runtime, Running a web application.		
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Textbooks

1	C# : A Beginners Guide – Childt , Herbert (Tata Mcgaw Hill , New Delhi)
2	C# The basics , Vijay Mukhi (BPB Publications)
3	C# Programming (Wrox Publications)
4	C# Programming Black Book – Matt Telles (DreamTech Publications)

Reference Books / Journal Articles / Weblink

1	C# Programming (Wrox Publications)
2	C# Programming Black Book – Matt Telles (DreamTech Publications)

MSCCS24308WD C# .NET Programming Lab

Course Type	DSE	Semester	III
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Teaching Scheme		Credits		Examination Scheme	
Lecture	03 Hr./Week	Lecture	0	CIE Marks	15
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	10
Practical/Studio	00 Hr./Week	Practical/Studio	2	Total Marks	25
Total	03 Hr./Week	Total	1		

COURSE OUTCOMES (COs)

CO1	Display proficiency in C# by building stand-alone applications in the .NET framework using C#.
CO2	Create distributed data-driven applications using the .NET Framework, C#, SQL Server and ADO.NET
CO3	Create web-based distributed applications using C#, ASP.NET, SQL Server and ADO.NET
CO4	Utilize DirectX libraries in the .NET environment to implement 2D and 3D animations and game-related graphic displays and audio.
CO5	Utilize XML in the .NET environment to create Web Service-based applications and components.
CO6	Understand the concept of Web Applications.

MAPPING BETWEEN COURSE OUTCOMES & PROGRAM OUTCOMES

(3/2/1 INDICATES STRENGTH OF CORRELATION) 3- High, 2- Medium, 1-Low

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	3	3	3	3	3	2	3								1, 2,3
CO2	3	3	3	3	3	3	3								3,4
CO3	3	3	3	3	3	3	3								3,4
CO4	3	3	3	3	3	3	3								4,5

CO5	2		2	2	3									3,4
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LIST OF EXPERIMENTS

1. Classes and objects
2. Inheritance
3. Operator overloading
4. Threading
5. Events and delegates
6. Working with windows forms controls
7. Validating data
8. Creating custom dialog box
9. Designing an MDI application with menu
10. Retrieving data from a SQL database
11. Manipulating data in a connected environment
12. Manipulating data in a disconnected environment